Demand Response Event Analysis: 6/14/22 Heat Wave

Customer Performance, Baseline Comparison, and Financial Results

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Event Overview & Methodology

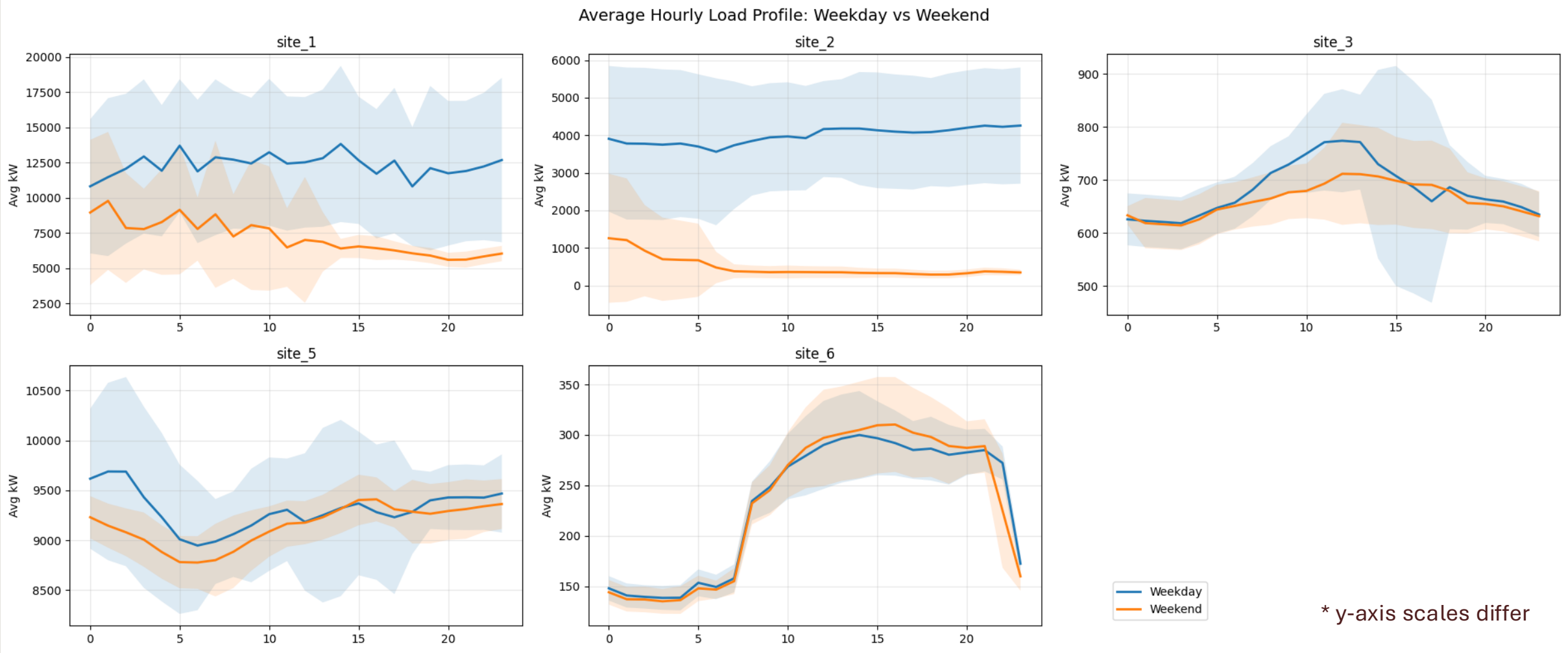
June 14, 2022 | 2:00–6:00 PM
Severe Midwest heat wave
High demand + low wind output
Emergency DR dispatch (first in ~6 months)

DATA
5 customer sites
15-min interval meter data (kWh), resampled to hourly kW.
Time-zone normalized to EDT

BASELINE

10of10	Firm Service Level (FSL)
Site-specific operational baseline	Fixed system benchmark per site
Average load during the 10 most recent comparable weekdays (or weekends), at the same hour	Based on MISO system peak conditions, independent of recent customer operations.

Portfolio Load Profiles



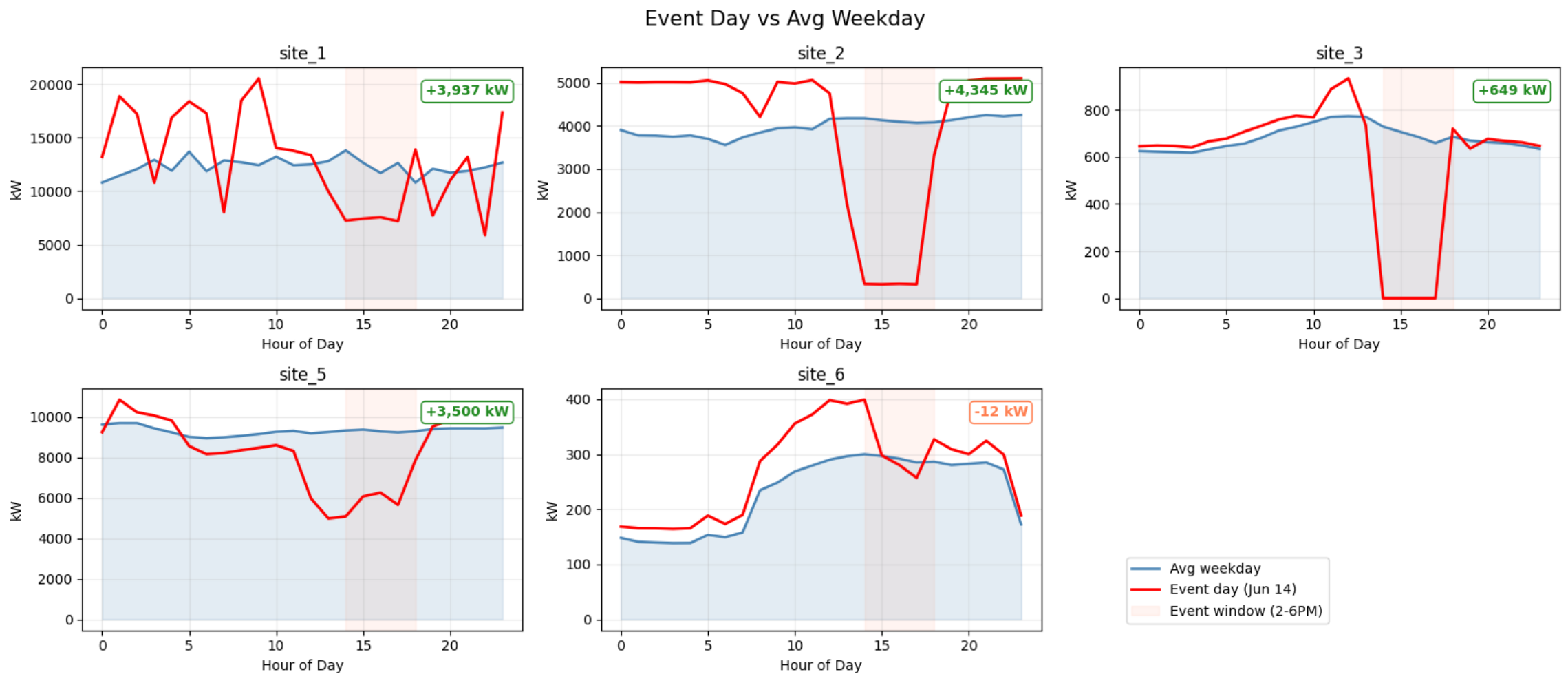
* y-axis scales differ

Building Type Hypothesis

Site	Likely Type	Daily Variability	Weekend Effect	Other Evidence
1	Large Industrial(M-F)	Large hourly swings (5MW -> 20+MW)	Moderate	Lower and more stable weekend load. Likely energy-intensive manufacturing processes that spikes up the load.
2	Mid-Sized Manufacturing (M-F)	Low	Large	Weekday mean load ~ 5MW with low volatility; nearly shut down on weekends.
3	Office/Institutional Buildings	Midday Peak	Minimal	Stable base load with low peak-base ratio (~1.6)
5	24/7 Continuous Load (e.g. mall, hospital)	Low, slight overnight trough	Minimal	Relatively stable load throughout; low peak-base ratio (~1.25)
6	Mid-sized business-hour building	Daily Cycle, high usage 8am-9pm	None	Low average demand (~200kW).

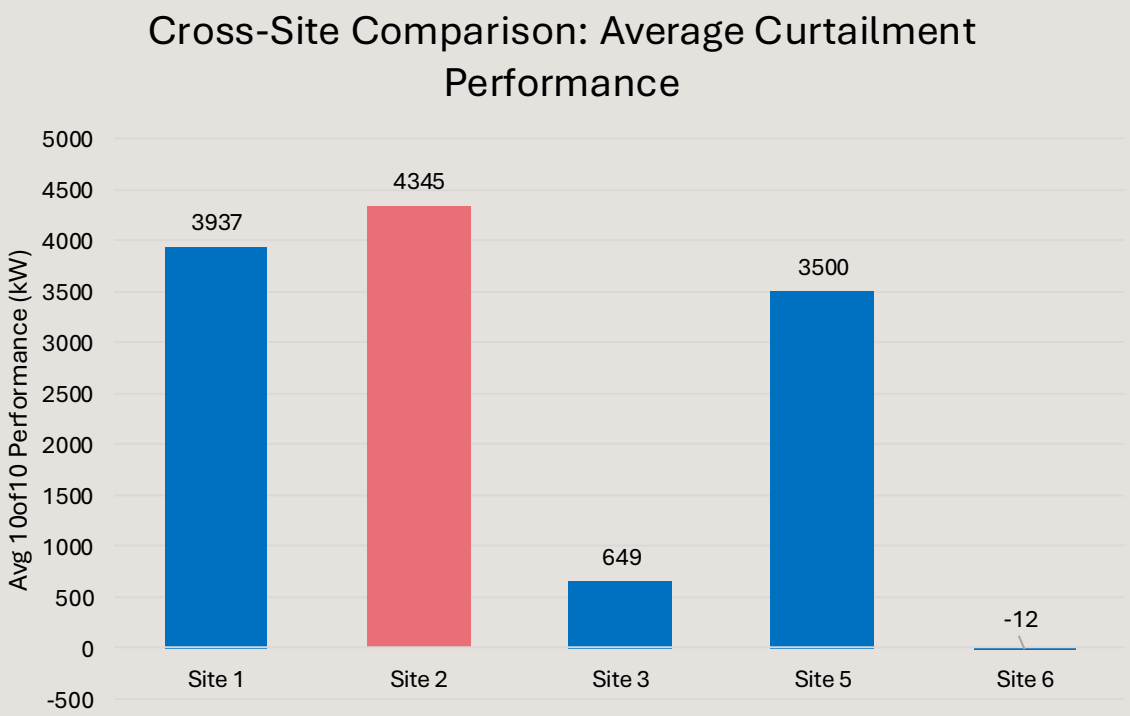
Event Response

Most sites significantly curtailed load during the June 14 event



Baseline Performance

Site	10of10 Avg (kW)	FSL Avg (kW)	Higher Baseline
Site 1	**3937**	3337	10of10
Site 2	4345	**5075**	FSL
Site 3	649	**850**	FSL
Site 5	**3500**	3235	10of10
Site 6	-12*	**41**	FSL

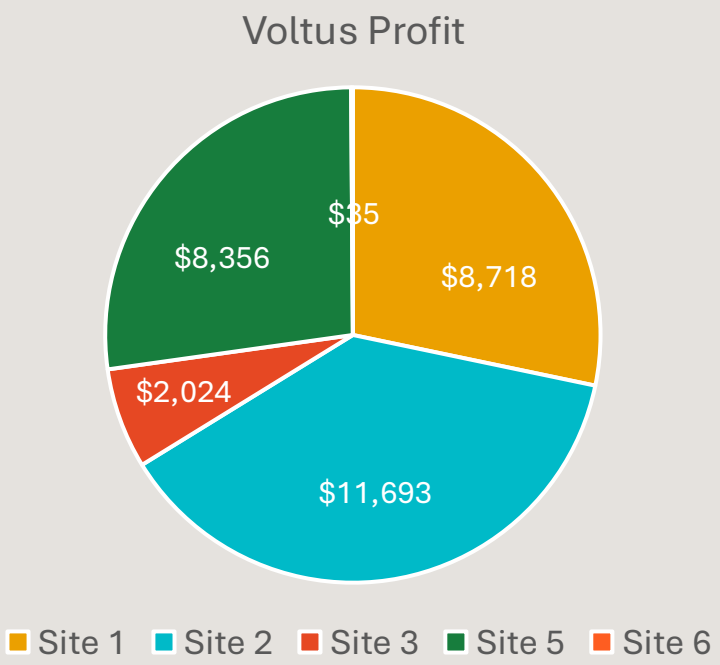


- Performance = Baseline – Actual, averaged over event hours
- Baseline methodology affects performance measured, and thus financial payout.
- Different operational pattern benefit from different baselines: if recent load was elevated, baseline rises, which can increase measured curtailment during the event.

* Negative performance indicates load above baseline. Treated as 0 for payout calculations per hour.

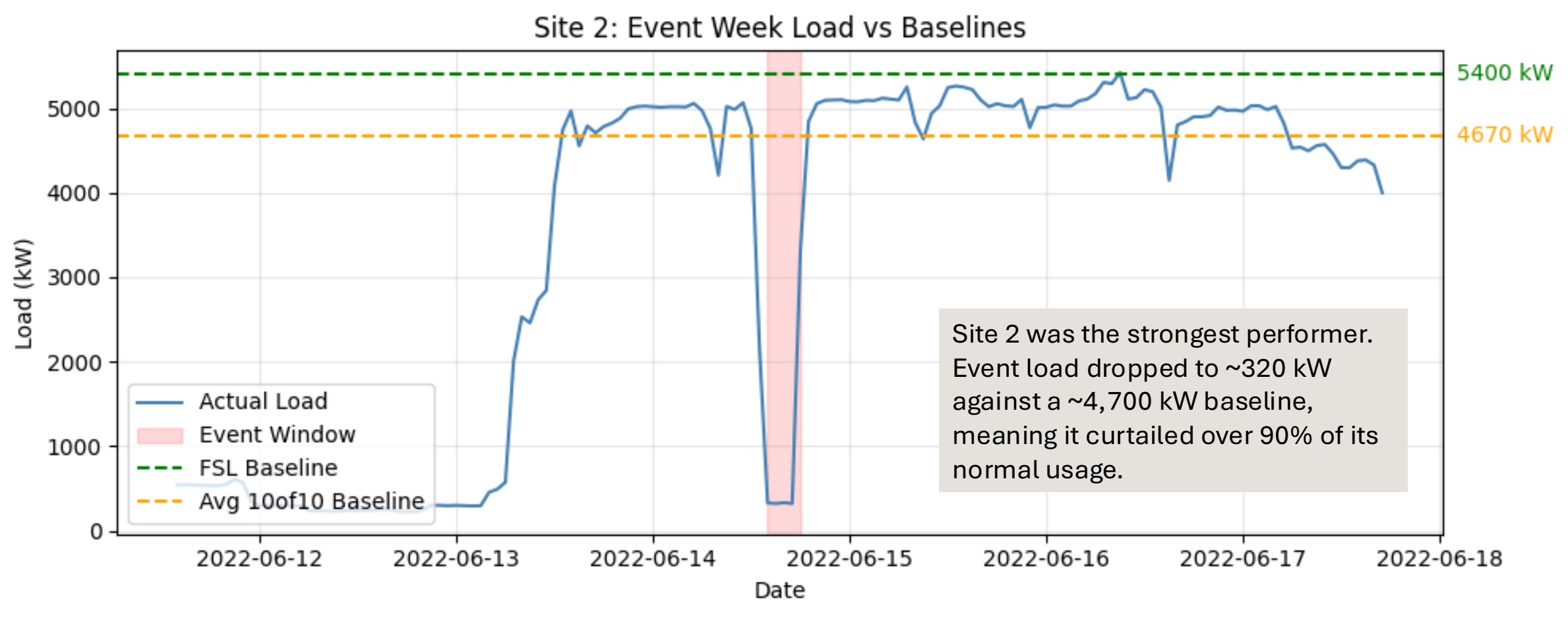
Financial Results

Site	Gross Revenue	Customer Payment	Voltus Profit
Site 1	$24,217	$15,499	$8,718
Site 2	$30,770	$19,077	$11,693
Site 3	$4,600	$2,576	$2,024
Site 5	$23,873	$15,518	$8,356
Site 6*	$71	$36	$35
Sum	$83,531	$52,706	$30,826

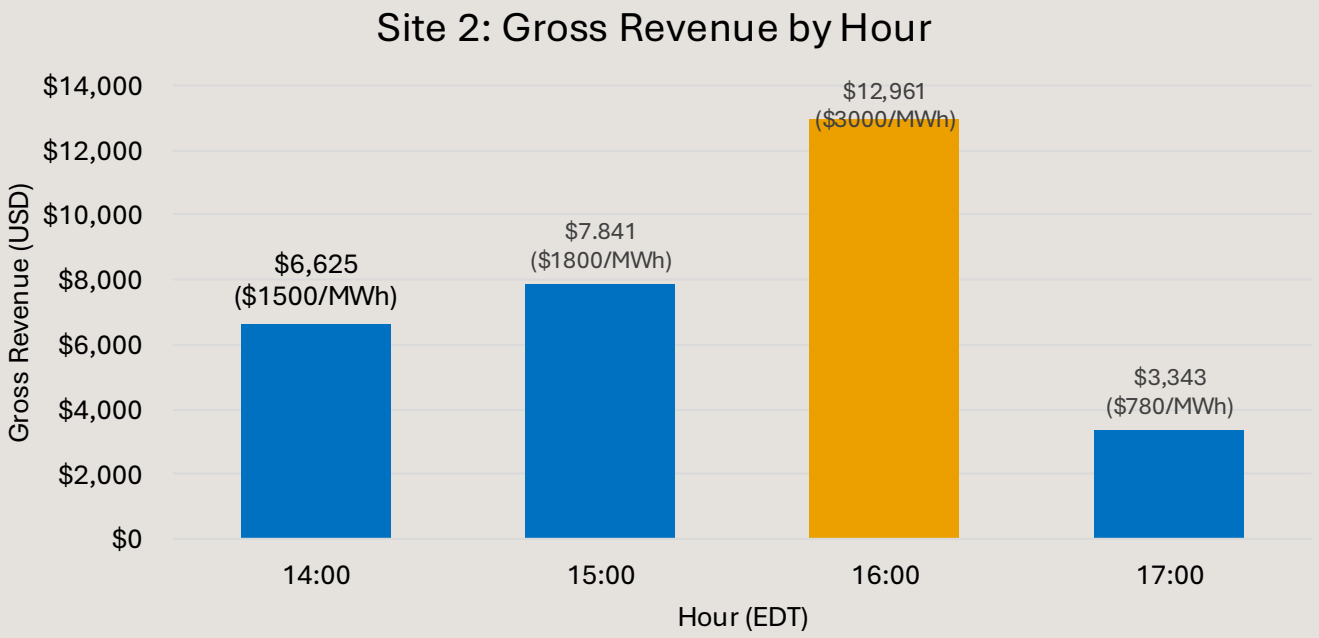


- Computed using the 10of10 baseline and performance.
- Total portfolio profit: $30.8K across 5 sites.
- Sites 1, 2, and 5 drive ~95% of total profit.

Site 2 Deep Dive



Site 2 was the strongest performer. Event load dropped to ~320 kW against a ~4,700 kW baseline, meaning it curtailed over 90% of its normal usage.



The 16:00 hour alone generated nearly $13K in gross revenue due to the $3,000/MWh LMP spike.

Hour	Load (kW)	10of10 Baseline (kW)	10of10 Performance (kW)	LMP ($/MWh)	Gross Rev	COGS	Profit
14:00	328	4,744	4,417	1,500	$6,625	$4,108	$2,518
15:00	320	4,676	4,356	1,800	$7,841	$4,861	$2,980
16:00	331	4,652	4,320	3,000	$12,961	$8,036	$4,925
17:00	320	4,606	4,286	780	$3,343	$2,073	$1,270
Total					**$30,770**	**$19,078**	**$11,693**

Scalability & Discussion

Q: If given 500 CSVs?

- Modular code with reusable functions.
- Runtime scale almost linearly with number of sites.
- Primary incremental effort: QA / validation
- Optionally, parallelize computations for each site

Strength of the Analysis

- Scalable for larger dataset; clear separation of logic
- Parameter-driven config (no hard-coded values)
- Clearly documented assumptions

Future improvement:

- Integrate holiday calendar and exclude them in weekday baseline calculations.
- Enhance ways to detect missing values/outliers/duplicates.
- Better handle edge cases: multi-day dispatch event, discontinuous event windows, etc.